

19 Implementation of Naive Bayes classification for Bank Loan prediction.

CODING:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

data = {

    'Age': [25, 30, 35, 40, 45, 28, 32, 38, 50, 29,

            42, 36, 27, 48, 34, 31, 44, 39, 26, 52],

    'Income': [25000, 35000, 45000, 60000, 75000, 30000, 40000, 55000,

               80000, 32000, 65000, 48000, 28000, 85000, 42000, 38000,

               70000, 58000, 27000, 90000],

    'LoanAmount': [100000, 150000, 200000, 250000, 300000, 120000,

                   180000, 220000, 350000, 140000, 280000, 210000,

                   110000, 320000, 190000, 160000, 270000, 230000,

                   100000, 400000],

    'CreditScore': [650, 680, 700, 750, 780, 660, 690, 720, 800, 670,

                    760, 710, 640, 790, 700, 680, 770, 730, 630, 810],

    'LoanStatus': [0, 0, 1, 1, 1, 0, 1, 1, 1, 0,

                   1, 1, 0, 1, 1, 0, 1, 1, 0, 1]

}

df = pd.DataFrame(data)

print("Bank Loan Dataset")

print("-----")

print(df)

X = df[['Age', 'Income', 'LoanAmount', 'CreditScore']]

y = df['LoanStatus']
```

```

X_train, X_test, y_train, y_test = train_test_split( X, y,
    test_size=0.30,
    random_state=42,
    stratify=y)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nNaive Bayes Classification Results")
print("-----")
print("Actual Values :", y_test.values)
print("Predicted Values:", y_pred)
print("Accuracy    :", accuracy)
print("Accuracy (%)  :", accuracy * 100)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=['Loan Rejected', 'Loan Approved']))
new_applicant = [[35, 50000, 180000, 720]]
prediction = model.predict(new_applicant)
print("\nNew Applicant Details")
print("-----")
print("Age      :", new_applicant[0][0])
print("Income   :", new_applicant[0][1])
print("Loan Amount :", new_applicant[0][2])

```

```
print("Credit Score:", new_applicant[0][3])
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if prediction[0] == 1:
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    print("Loan Prediction: APPROVED")
```

```
else:
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```
    print("Loan Prediction: REJECTED")
```

OUTPUT:

Bank Loan Dataset					
	Age	Income	LoanAmount	CreditScore	LoanStatus
0	25	25000	100000	650	0
1	30	35000	150000	680	0
2	35	45000	200000	700	1
3	40	60000	250000	750	1
4	45	75000	300000	780	1
5	28	30000	120000	660	0
6	32	40000	180000	690	1
7	38	55000	220000	720	1
8	50	80000	350000	800	1
9	29	32000	140000	670	0
10	42	65000	280000	760	1
11	36	48000	210000	710	1
12	27	28000	110000	640	0
13	48	85000	320000	790	1
14	34	42000	190000	700	1
15	31	38000	160000	680	0
16	44	70000	270000	770	1
17	39	58000	230000	730	1
18	26	27000	100000	630	0
19	52	90000	400000	810	1

Naive Bayes Classification Results	
Actual Values	: [0 1 0 1 1 1]
Predicted Values	: [0 1 0 1 1 1]
Accuracy	: 1.0
Accuracy (%)	: 100.0

Confusion Matrix:	
[[2 0]	
[0 4]]	

Classification Report:				
	precision	recall	f1-score	support
Loan Rejected	1.00	1.00	1.00	2
Loan Approved	1.00	1.00	1.00	4
accuracy			1.00	6
macro avg	1.00	1.00	1.00	6
weighted avg	1.00	1.00	1.00	6

New Applicant Details

Age : 35

Income : 50000

Loan Amount : 180000

Credit Score: 720

Loan Prediction: APPROVED